

Data Types, Variables & Control Flow

1. Primitive Data Types

Java is a strongly typed language, which means that every variable must be declared with a data type. There are 8 primitive data types in Java:

- **byte:** 8-bit signed two's complement integer. (min -128, max 127)
- **short:** 16-bit signed two's complement integer.
- **int:** 32-bit signed two's complement integer.
- **long:** 64-bit signed two's complement integer.
- **float:** single-precision 32-bit IEEE 754 floating point.
- **double:** double-precision 64-bit IEEE 754 floating point.
- **boolean:** has only two possible values: true and false.
- **char:** single 16-bit Unicode character.

2. Type Casting

Type casting is when you assign a value of one primitive data type to another type.

- **Widening Casting (Automatic):** converting a smaller type to a larger type size. byte
-> short -> char -> int -> long -> float -> double
- **Narrowing Casting (Manual):** converting a larger type to a smaller size type. double
-> float -> long -> int -> char -> short -> byte

3. Control Flow: If-Else and Switch

Java uses control flow statements to control the flow of execution of a program based on certain conditions.

If-Else Statement: executes a block of code if a specified condition is true. If the condition is false, another block of code can be executed.

Switch Statement: allows a variable to be tested for equality against a list of values. Each value is called a case, and the variable being switched on is checked for each case.

4. Loops in Java

- **for loop:** Useful when you know exactly how many times you want to loop through a block of code.
- **while loop:** Loops through a block of code as long as a specified condition is true.
- **do-while loop:** This loop will execute the code block once, before checking if the condition is true, then it will repeat the loop as long as the condition is true.

5. Break and Continue

The **break** statement can also be used to jump out of a loop.

The **continue** statement breaks one iteration (in the loop), if a specified condition occurs, and continues with the next iteration in the loop.